

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0117

Southwyck 138 kV
January 28, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0117. The Interconnection Customer proposes a 398.4 MW natural gas combined cycle facility interconnecting at the Southwyck 138 kV substation in the Kentucky area. The request was received on March 21, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 82 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$58,688,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CAVANAUGH / HIDROELECTRICA DOS BOCAS 8005 138 KV
Contingency: SOUTHWYCK / BLUE MTN 138 KV
Loading: 203.2% (Rating: 188 MVA)
DFAX: 100.0%
Pre-project loading: 9.1%
2. Facility: BLUE MTN SUBSTATION / SOUTHWYCK 138 KV
Contingency: CAVANAUGH / SOUTHWYCK 138 KV L
Loading: 194.5% (Rating: 200 MVA)
DFAX: 100.0%
Pre-project loading: 4.3%
3. Facility: SOUTHWYCK / CAVANAUGH 138 KV
Contingency: BLUE MTN / SOUTHWYCK 138 KV
Loading: 187.4% (Rating: 208 MVA)
DFAX: 100.0%
Pre-project loading: 4.2%
4. Facility: BLUE MOUNTAIN / LANK 138 KV
Contingency: CAVANAUGH / SOUTHWYCK 138 KV
Loading: 186.7% (Rating: 205 MVA)
DFAX: 100.0%
Pre-project loading: 7.7%
5. Facility: HIDROELECTRICA DOS BOCAS 8005 / WANSLEY COMBINED CYCLE 138 KV
Contingency: BUE MTN / SOUTHWYCK 138 KV Line
Loading: 181.6% (Rating: 202 MVA)
DFAX: 100.0%
Pre-project loading: 15.4%

6. Facility: SPRING CREEK RD / HARRIS 138 KV line
 Contingency: BLUE MOUNTAIN SUBSTATION / SOUTHWYCK 138 KV
 Loading: 148.3% (Rating: 185 MVA)
 DFAX: 22.9%
 Pre-project loading: 98.9%
7. Facility: ATOKA / SPRING CREEK ROAD 138 KV
 Contingency: WANSLEY COMBINED CYCLE / SCOTT BRANCH 138 KV Line
 Loading: 109.2% (Rating: 228 MVA)
 DFAX: 32.2%
 Pre-project loading: 52.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CAVANAUGH / HIDROELECTRICA DOS BOCAS 8005 138 KV	Rebuild line with higher-capacity conductor	\$7,281,000	100% (\$7,281,000)
BLUE MOUNTAIN / SOUTHWYCK 138 KV	Rebuild line with higher-capacity conductor	\$30,467,000	100% (\$30,467,000)
SOUTHWYCK / CAVANAUGH 138 KV	Rebuild line with higher-capacity conductor	\$7,796,000	100% (\$7,796,000)
BLUE MOUNTAIN / LANK 138 KV	Rebuild line with higher-capacity conductor	\$1,733,000	100% (\$1,733,000)
WANSLEY COMBINED CYCLE / HIDROELECTRICA DOS BOCAS 8005 138 KV line	Rebuild line with higher-capacity conductor	\$2,212,000	100% (\$2,212,000)
SPRING CREEK RD / HARRIS 138 KV line	Rebuild line with higher-capacity conductor	\$3,457,000	95% (\$3,294,000)
ATOKA / SPRING CREEK ROAD 138 KV	Reconductor line	\$6,085,000	97% (\$5,905,000)
	Total allocated to project		\$58,688,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.